Retail Sales EDA Project Report

Oasis Infobyte Internship (OIBSIP) · Data Analytics Track · Task 1

Analyst: Gajanan Sanjay Kalbande

1. Executive Summary

This exploratory data analysis (EDA) report examines 1,000 retail transaction records to evaluate customer purchasing habits, time-series revenue trends, and product performance. Using Python data libraries (Pandas, NumPy, Seaborn), the analysis transforms raw transactional data into actionable commercial intelligence.

TOTAL RECORDS	MISSING VALUES	OUTLIERS (IQR)	TOP REVENUE
1,000 Rows	0 (Clean)	0 Found	Electronics

2. Dataset Structure & Schema

Column Name	Data Type	Description
Transaction ID	Integer (int64)	Unique numerical identifier for each transaction (1 to 1,000).
Date	Datetime / Object	Calendar date when the transaction occurred.
Customer ID	String	Unique alphanumeric code assigned to each customer.
Gender	String	Recorded customer gender (Male / Female).
Age	Integer (int64)	Customer age ranging from 18 to 64 years.
Product Category	String	Category of item purchased (Beauty, Clothing, Electronics).

3. Statistical Findings & Outlier Analysis

Descriptive statistics reveal a right-skewed spend distribution where the mean transaction amount ($456.00) exceeds the median ($135.00). Applying the Interquartile Range (IQR) method (Q1 = $60, Q3 = $900, IQR = $840) established an upper bound threshold of $2,160. Because the maximum recorded transaction is $2,000, **zero statistical outliers** exist, confirming that high expenditures reflect legitimate premium product pricing rather than data errors.

4. Strategic Business Recommendations

- **Inventory Allocation:** Scale up stock levels for high-revenue categories like Electronics ahead of peak seasonal demand periods.
- **Cross-Selling Bundles:** Leverage high-frequency unit movement in Beauty and Clothing by introducing multi-item product bundles to increase average basket size.
- **Broad-Market Marketing:** Maintain even marketing allocations across demographic cohorts, as age brackets (25–65) and gender splits are well-balanced.

5. Project Conclusion

This analysis successfully converted transactional records into robust commercial intelligence. Confirmed by rigorous IQR outlier checks and stable time-series trends, the findings demonstrate consistent consumer demand across balanced demographic segments, with Electronics acting as the primary revenue catalyst.

> **Key Takeaway:** High-value transactions are driven entirely by premium product pricing. Strategic inventory and bundling initiatives provide a clear roadmap for maximizing profitability.